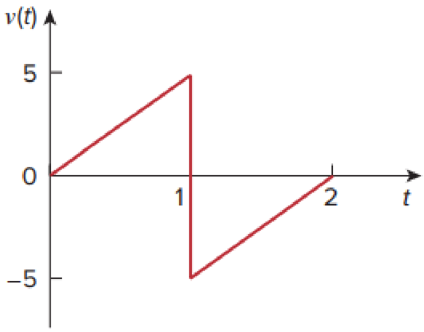
Chapter 6, Problem 45P

Problem

If the voltage waveform in Fig. 6.68 is applied to a 25-mH inductor, find the inductor current *i*(*t*) for 0<*t*<2 seconds. Assume *i*(0) = 0.

Fig. 6.68:



Step-by-step solution

Step 1 of 3

Refer to voltage waveform in Figure 6.68 in the textbook.

Calculate the voltage equation for $0 < t < 1\,\text{sec}$.

For $0 < t < 1\,\text{sec}$, points are (0,0) (1,5)

Calculate the voltage equation.

$$(v-0) = \left(\frac{5-0}{1-0}\right)(t-0)$$
$$v = 5t \text{ V}$$

Calculate the voltage equation for $1 < t < 2\,\text{sec}$.

For $1 < t < 2\,\text{sec}$, points are (1,-5) (2,0)

$$(v-(-5)) = \left(\frac{0-(-5)}{2-1}\right)(t-1)$$
$$v+5 = 5t-5$$
$$v = 5t-10 \text{ V}$$

So Inductor voltage is

$$v = \begin{cases} 5t \text{ V} & 0 < t < 1 \\ 5t-10 \text{ V} & 1 < t < 2 \end{cases}$$

Step 2 of 3

Inductor current at $t = 0$ is zero.

$$i(0) = 0 \text{ A}$$

Calculate the current expression for $0 < t < 1 \text{ sec}$.

$$\begin{aligned}
i &= \frac{1}{L}\int_0^t v(t)\,dt + i(0) \\
&= \frac{1}{25\times10^{-3}}\int_0^t 5t\,dt \\
&= \left(\frac{5}{25\times10^{-3}}\right)\frac{t^2}{2} \\
&= 100t^2 \text{ A}
\end{aligned}$$

Calculate the current at 1 s.

$$\begin{aligned}
i(1\text{ s}) &= 100(1)^2 \\
&= 100 \text{ A}
\end{aligned}$$

Step 3 of 3

Calculate the current expression for $1 < t < 2 \text{ sec}$.

$$\begin{aligned}
i &= \frac{1}{L}\int_1^t v(t)\,dt + i(1) \\
&= \frac{1}{25\times10^{-3}}\int_1^t (5t-10)\,dt + 100 \\
&= 40\left(5\frac{t^2}{2} - 10t\right)\Bigg|_1^t + 100 \\
&= 40\left[2.5t^2 - 10t - (2.5-10)\right] + 100 \\
&= 40\left[2.5t^2 - 10t + 7.5\right] + 100 \\
&= 100t^2 - 400t + 300 + 100 \\
&= 100t^2 - 400t + 400 \text{ A}
\end{aligned}$$

Therefore, the inductor current for $0 < t < 2\text{sec}$ is,

$$\boxed{i(t) = \begin{cases} 100t^2 \text{ A} & 0 < t < 1 \text{ sec} \\ 100t^2 - 400t + 400 \text{ A} & 1 < t < 2 \text{ sec} \end{cases}}.$$